

STEVEN COMPÈRE

PhD Materials Chemist & Chemical Engineer

Carbon Nanotubes · CVD Process Development · Materials Characterization · R&D Project Management
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PROFESSIONAL SUMMARY

Chemical engineer and PhD materials scientist with 5+ years of R&D in academic and industrial settings. Currently growing vertically aligned carbon nanotubes by aerosol-assisted catalytic CVD at CEA Paris-Saclay, after a PhD on the CVD synthesis of nanostructured carbons (1 patent application, 6 peer-reviewed articles). Combines hands-on process development and multi-technique characterization (SEM/TEM, Raman, physisorption, TGA, XPS, XRD) with GMP-grade documentation, Python data analysis, R&D project management and team training.

CORE SKILLS

CNT & CVD processes	Aerosol-assisted catalytic CVD (CCVD) of vertically aligned CNTs; ethylene CVD up to 1000 °C; hybrid solid-state/CVD routes; gas flow control; process optimization.
Experimental setups	Modification, maintenance and repair of setups; reactor concept design; high-pressure gas adsorption and deformation measurements; RIE plasma etching.
Characterization	SEM/EDS, TEM, Raman, XPS, XRD, FT-IR, UV-Vis, EPR, solid-state NMR, TGA/DSC, physisorption (N ₂ , Ar, CO ₂), Hg porosimetry, He pycnometry, CHNS, ICP-OES.
Electrical analysis	Electrical conductivity by impedance spectroscopy; electrostatic charge measurements on an in-house setup.
Data & software	Python (data processing, IAST modeling, AI-assisted coding; author of the open-source AdsorpSuite GUI, github.com/stevencompere/AdsorpSuite), OriginLab, ImageJ, SIMCA (PCA), digital image correlation.
Quality & GMP	GMP/GLP, analytical method development and validation, traceability, 5S, Lean, HSE, GMP cleanroom practices.
Leadership	R&D project management, training and coordination of colleagues, intern supervision, cross-functional collaboration, scientific communication.
Languages	French (native), English (C1).

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher – Carbon Nanotube Growth & Membranes CEA Paris-Saclay, NIMBE/LEDNA laboratory	06/2026 – Present Gif-sur-Yvette, France
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- **ANR CANELONI project** (nanoporous membranes based on vertically aligned CNTs): grow VACNT carpets by aerosol-assisted catalytic CVD, setting process parameters, precursors and catalyst feed (ferrocene).
- Characterize CNT carpets by SEM, TEM, Raman and TGA, and interpret quality, CNT diameter, density and alignment to steer growth conditions.
- Fabricate VACNT-based nanoporous membranes, including RIE plasma etching; propose new operating procedures and research directions (e.g., sulfur as a growth promoter, zeolite-based nanostructured substrates).

PhD Researcher – Materials Chemistry IC2MP (CNRS / Université de Poitiers) & LFCR (Université de Pau)	10/2022 – 10/2025 Poitiers & Anglet, France
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- Developed and optimized ethylene CVD processes (up to 1000 °C) and hybrid solid-state/CVD co-feeding routes to control the structure and porosity of zeolite-templated carbons (ZTC) for energy and gas capture.
- Modified, maintained and repaired gas-phase CVD setups; developed the conceptual design of a rotary reactor for mg-to-g scale-up of ZTC synthesis (process flow, equipment specifications, operating windows).
- Synthesized a new lamellar-type microporous carbon, leading to a patent application filed with Yokohama National University (Japan, 2025).
- Established structure–property relationships (zeolite acidity, counter-ion, synthesis temperature) by combining XRD, Raman, XPS, physisorption, solid-state NMR and EPR; 6 peer-reviewed articles, 3 as first author.
- Developed a shaping route for robust ZTC pellets; measured high-pressure adsorption (CO₂, CH₄, H₂) and adsorption-induced deformation by digital image correlation during a 3-month mission at LFCR.
- Conducted characterization campaigns at CEMHTI (Orléans, solid-state NMR) and LASIRE (Lille, EPR); fully trained and supervised a Master's (M1) intern for 3 months.

R&D Engineer – Materials Science

UCB Pharma (consultant with Jefferson Wells)

09/2020 – 08/2022

Braine-l'Alleud, Belgium

- Managed physical characterization projects in a GMP/GLP environment, from experimental design and analytical method development/validation to data analysis and reporting.
- Operated a broad powder-characterization toolbox: FT4 powder rheometer, Schulze ring shear tester, Granutools, laser diffraction, morphogranulometry, DVS, tapped density, XRD, SEM, optical microscopy.
- Trained and coordinated colleagues on characterization methods and instruments.
- Partnered with Formulation, Process and Production teams to troubleshoot industrial issues and support innovation, including interventions in GMP cleanroom areas.
- Implemented 5S in the sample library (5,000+ samples), ensuring full traceability and inventory integrity.

Materials Science Intern (Master's thesis)

UCB Pharma

09/2019 – 12/2019

Braine-l'Alleud, Belgium

- Built a reference database of excipient properties and applied multivariate analysis (PCA) to identify redundant tests, improving the efficiency of characterization protocols.

Analytical Chemistry Intern (Bachelor's thesis)

TotalEnergies

01/2017 – 05/2017

Feluy, Belgium

- Developed a diffuse-reflectance UV-Vis protocol for synthesized silica-supported metallocene catalysts to compare their activation states.

EDUCATION & TRAINING

PhD in Materials Chemistry

Université de Poitiers (IC2MP)

2022 – 2025

Poitiers, France

Thesis: Zeolite-templated carbons: synthesis, functionalization, shaping and high-pressure gas adsorption-induced deformation (defended 10/2025).

Master of Engineering in Industrial Chemistry (Ingénieur industriel)

Haute École Louvain en Hainaut (HELHa)

2017 – 2020

Belgium

BSc in Applied Chemistry

Haute École Louvain en Hainaut (HELHa)

2014 – 2017

Belgium

Transmission electron microscopy (TEM) training, LPICM, École polytechnique, France

2026

Coursework toward an MSc in Management Sciences (evening program), UCLouvain, Belgium

2021 – 2022

LEADERSHIP & ENGAGEMENT

- Elected representative of non-permanent staff on the IC2MP Laboratory Council (2022–2023), taking part in strategic and budget decisions.
- Organized scientific events for PhD students within the Picta'Sciences association (Université de Poitiers).

PUBLICATIONS & PATENT

- **Patent application:** S. Compère et al., "Porous carbon-based material and method for producing porous carbon-based material," Japan, filed 09/2025 (with Yokohama National University).
- S. Compère et al., "Functionalized zeolite-templated carbons via solid-gas co-feeding synthesis," *Microporous Mesoporous Mater.* 417 (2027) 114414.
- S. Compère, T. Aumond, I. Batonneau-Gener, H. Vezin, V. Montouillout, A. Sachse, "Influence of the zeolite counterion on the properties of zeolite-templated carbons," *Microporous Mesoporous Mater.* 407 (2026) 114107.
- S. Compère, R. Minhas, I. Batonneau-Gener, C. Miqueu, L. Perrier, A. Sachse, "Zeolite-templated carbon pellets by in situ generated binder: deformation behavior under high-pressure CO₂ adsorption," *Microporous Mesoporous Mater.* 399 (2026) 113837.
- L. Loupias, ..., S. Compère, ..., A. Sachse, A. Habrioux, "N,S-doped zeolite-templated carbon like supported Ni_xFe_yS_z as an efficient bifunctional catalyst for rechargeable zinc-air batteries," *ChemCatChem* 16 (2024) e202400555.
- T. Aumond, S. Compère, A. Sachse, "Impact of thermal ethylene degradation in zeolite-templated carbon synthesis," *Microporous Mesoporous Mater.* 367 (2024) 112967.
- T. Aumond, H. Vezin, I. Batonneau-Gener, S. Compère, Y. Pouilloux, A. Le Person, A. Moissette, A. Sachse, "Acidity: a key parameter in zeolite-templated carbon formation," *Small* 19 (2023) 2300972.
- Further first-author manuscripts in preparation, including a review. 8 conference contributions (5 oral), e.g., Fundamentals of Adsorption (Porto, 2025), FEZA (Portorož, 2023), ZMPC (Osaka, 2024). Full publication list: ORCID 0009-0006-3474-8748.